

Decision memo: the Vikarabad HPQ project, measurement programme before build

Prepared for the investment committee. Every number in this memo was produced by running the models in this repository during the analysis session that wrote it. Provenance tags are MEASURED (from a physical measurement), SOURCED (from a citable published price or standard), COMPUTED (from a model over other tagged inputs), DERIVED (arithmetic on SOURCED values plus a stated judgement), and ASSUMED (a scenario choice with no measurement behind it). Reproduction commands are listed in section 8. Every figure quoted below appears in docs/memo_numbers.csv, one row per quantity.

The single most important statement in this memo: the Vikarabad deposit is UNCHARACTERIZED in the citable record. No assay of it exists in any source consulted. Every ore parameter below is therefore an ASSUMED scenario band, not a property of the rock, and every conclusion is conditional on a future assay campaign. Nothing here licenses a build decision.

1. The decision on the table

Three actions are modelled. They are ASSUMED structure, not measurements.

Action	Content	Incremental capex	Incremental opex
walk_away	do not build, value 0 by definition	0	0
build_base	flotation plus hot acid leach, HPQ sand product	25 to 60 MUSD band	in cash cost
build_polish	base plant plus a polishing stage that cuts Al by a factor 0.60	+9.0 MUSD (ASSUMED)	+35.0 USD/t product (ASSUMED)

The polishing factor is bounded strictly between 0 and 1 on physical grounds: lattice-bound Al substitutes for Si in the quartz structure and is not removable by any surface or gas-phase treatment, so no polishing stage can drive Al to zero. That bound is asserted by a test (test_polish_action_cannot_remove_all_aluminium).

Under the prior input bands, expected values are walk_away 0.000, build_base +18.943, build_polish +11.168 MUSD (COMPUTED, 200,000 draws, seed 1234). The prior-best action is build_base. Per-draw, all three actions win somewhere: over 4,000 draws the argmax was walk_away 332 times, build_base 3,661 times, build_polish 7 times, so the decision is live rather than degenerate.

2. Which uncertainty dominates: Sobol decomposition

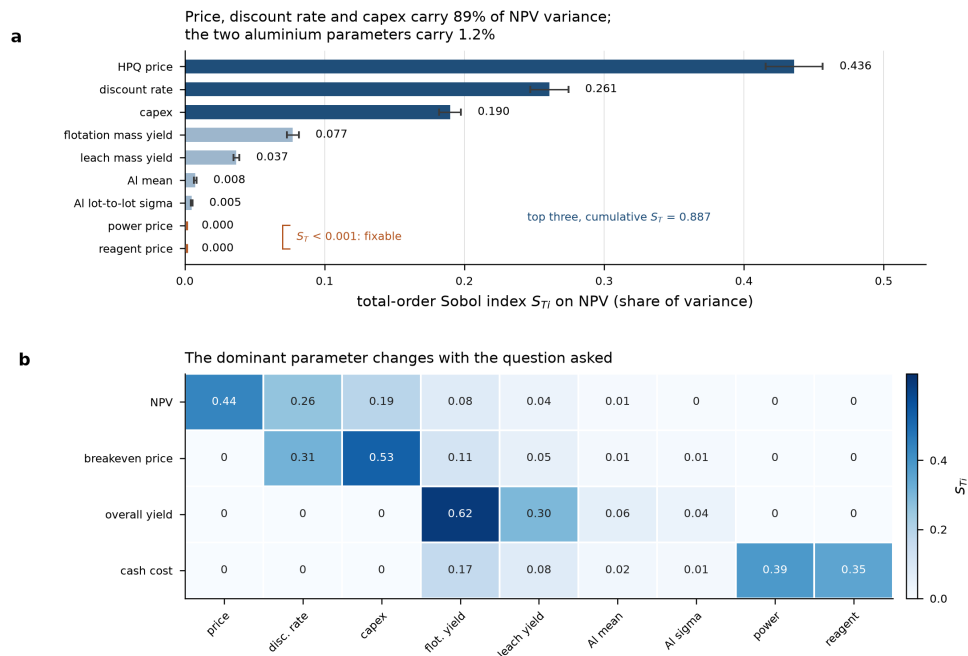
Variance-based global sensitivity, Saltelli scheme via SALib, over the integrated chain plant_npv in tests/test_integration_uncertainty.py. N = 4,096 base samples, 45,056 model evaluations, on an NPV output variance of 204.85 MUSD^2. All values COMPUTED.

Rank	Parameter	Total-order S_T	95% CI	First-order S_1
1	HPQ price	0.4361	+/- 0.0204	0.4256
2	discount rate	0.2607	+/- 0.0137	0.2508
3	capex	0.1897	+/- 0.0078	0.1898
4	flotation mass yield	0.0773	+/- 0.0042	0.0749
5	leach mass yield	0.0369	+/- 0.0022	0.0359
6	Al mean	0.0075	+/- 0.0009	0.0041
7	Al lot-to-lot sigma	0.0049	+/- 0.0006	0.0025
8	power price	0.0000	+/- 0.0000	0.0001
9	reagent price	0.0000	+/- 0.0000	0.0000

The three parameters that dominate NPV variance are HPQ price, discount rate, and capex, carrying a cumulative total-order index of 0.887. The model is essentially additive: the sum of first-order indices is 0.9837 of the sum of total-order indices, so interactions account for under 2 percent of variance and one-parameter-at-a-time reasoning is defensible here.

The result is stable. Re-running at N = 2,048 and N = 1,024 leaves the ranking unchanged with a maximum total-order drift of 0.0004, and price is the top input at every level.

The dominant parameter changes with the question asked (COMPUTED, N = 2,048, 22,528 evaluations each). On breakeven price, capex leads at 0.5264 and discount rate follows at 0.3104. On overall yield, flotation mass yield leads at 0.6226 and leach mass yield at 0.2968. On cash cost, power price leads at 0.3854 and reagent price at 0.3511, both of which are irrelevant to NPV. A memo that reported sensitivity on one output would mis-rank the others.



Source: this study, Saltelli scheme (Saltelli 2002; Saltelli et al. 2010) via SALib over plant_npv in tests/test_integration_uncertainty.py.
NPV panel N=4096 base (45,056 evaluations); other outputs N=2048 (22,528 each). Bars in (a) are SALib 95% bootstrap CI.
Input bands are ASSUMED scenario bands, not assays: the Vikarabad deposit is uncharacterized.

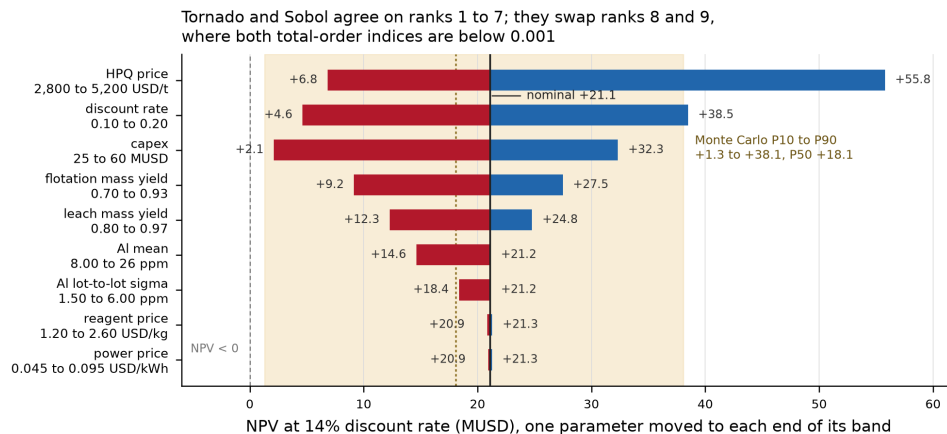
Figure 1. (a) Total-order Sobol indices on NPV with SALib 95 percent bootstrap confidence intervals; the top three are highlighted. (b) Total-order indices for four model outputs, showing that the dominant parameter is a property of the output chosen, not of the project. Source: this study, `scripts/memo_run.py`; Saltelli 2002 (doi:10.1016/S0010-4655(02)00280-1), Saltelli et al. 2010 (doi:10.1016/j.cpc.2009.09.018).

3. The tornado

One-at-a-time swings, each parameter moved to each end of its band with all others held at their triangular mode (power price has no mode, held at its uniform midpoint 0.07 USD/kWh). Nominal NPV at those modes is +21.109 MUSD (COMPUTED).

Parameter	Band	NPV at low	NPV at high	Swing
HPQ price	2,800 to 5,200 USD/t	+6.82	+55.77	48.95
discount rate	0.10 to 0.20	+38.48	+4.64	33.84
capex	25 to 60 MUSD	+32.33	+2.11	30.23
flotation mass yield	0.70 to 0.93	+9.16	+27.48	18.33
leach mass yield	0.80 to 0.97	+12.28	+24.79	12.51
Al mean	8 to 26 ppm	+21.16	+14.62	6.53
Al lot-to-lot sigma	1.5 to 6.0 ppm	+21.16	+18.37	2.80
reagent price	1.20 to 2.60 USD/kg	+21.30	+20.86	0.44
power price	0.045 to 0.095 USD/kWh	+21.27	+20.95	0.33

The tornado and the Sobol decomposition agree on ranks 1 through 7 and swap only ranks 8 and 9, where both total-order indices are below 0.001. Two structural points the tornado makes that the Sobol table does not: no single parameter taken to its adverse end drives NPV negative (the worst single-parameter outcome is capex at 60 MUSD, +2.11 MUSD), and the Al parameters are one-sided, since Al only ever hurts (both low ends sit at the nominal, both high ends below it), which is the specification-yield cliff rather than a symmetric revenue effect.



Source: this study, One-at-a-time tornado over plant_npv, all other parameters held at their triangular mode; power price has no mode (uniform band), held at 0.07 USD/kWh. Shaded band is the 20,000-draw Monte Carlo P10 to P90, seed 42. Blue arm = NPV rises, red arm = NPV falls. Bands are ASSUMED, not assays.

Figure 2. One-at-a-time tornado on NPV, arms coloured by direction of effect, with the 20,000-draw Monte Carlo P10 to P90 band shaded behind for scale. Source: this study, `scripts/memo_run.py`.

4. What to measure first: the value of information

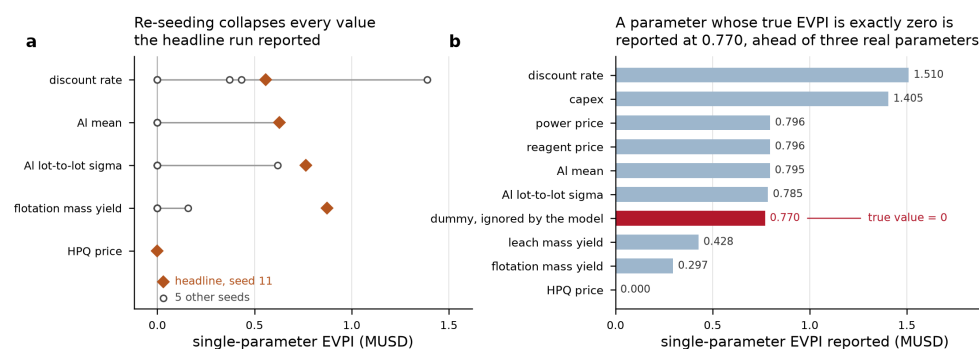
This is the recommendation section, and it carries a methodological finding that changes what can be recommended.

4.1 The per-parameter EVPI ranking is not reportable

The shipped estimator `ae.agent.decisions.evpi` returned an apparently clean ranking at the headline settings (`n_outer` = 256, `n_inner` = 256, seed 11): flotation mass yield 0.8731, AI sigma 0.7640, AI mean 0.6292, discount rate 0.5566, power price 0.0329 MUSD, with four parameters at exactly 0.0000. Three controls show that ranking is an artefact.

- Switch fraction.** EVPI is the value of changing your decision. The estimator reported the recommended action changing on exactly 0.000 of draws for all nine parameters at every seed tested. A positive EVPI alongside a zero switch fraction is arithmetic on noise.
- Seed replication.** Re-running five parameters at six seeds (11, 23, 37, 53, 71, 97) collapsed four of five to exactly 0.0000 at five of six seeds. The estimator forms its own baseline per call, and that baseline has a standard deviation of 0.6776 MUSD across seeds, which is larger than every EVPI it reports.
- Null calibration.** A dummy parameter drawn from a prior and then discarded by the value function has a true EVPI of exactly zero by construction. The shipped estimator reported it at 0.7698 MUSD, placing it ahead of three real parameters.

Under a consistent shared baseline (65,536 draws) the dummy's floor falls to +0.1233 MUSD. Six parameters clear that floor, two of them substantially (discount rate +0.8635, capex +0.7588), so those two are plausibly non-zero. But of the four parameters an assay campaign can actually resolve, the two AI parameters sit only marginally above the floor (+0.1488 and +0.1387, margins of 21 and 12 percent of the floor) and the two mass yields come out NEGATIVE (-0.3498, -0.2186), which is only possible as sampling error since EVPI is non-negative by construction. **No ordering of the assay-resolvable parameters survives its own noise, so this memo does not publish one.**



Source: this study, `scripts/memo_evpi_control.py`. (a) `ae.agent.decisions.evpi` at `n_outer=256`, `n_inner=256`, seeds 11, 23, 37, 53, 71, 97. (b) same estimator at seed 11, with a dummy input drawn from a prior and discarded by the value function. The estimator forms its own baseline per call (sd 0.678 MUSD across the six seeds) and clamps the difference at zero, so baseline noise appears as one-sided positive EVPI. Switch fraction was exactly 0.000 for all nine real parameters at every seed tested.

Figure 3. (a) Seed replication: the headline value (diamond) against five other seeds (circles) for each of five parameters. (b) Null calibration: a parameter with true EVPI of exactly zero is reported at 0.770. Source: this study, `scripts/memo_evpi_control.py`.

4.2 Group EVPI, and the number that does resolve

An assay campaign resolves ore parameters together, not one at a time, so group EVPI is the better-posed question. Computed by definition with no clamping at zero and one shared baseline (131,072 draws), at `n_outer` = 512, `n_inner` = 512 (COMPUTED):

Group	Cost, USD	EVPI, MUSD	95% CI	Switch fraction
all nine, nested	215,478	+0.1793	-1.025 to +1.383	0.094
commercial, all five	213,000	+0.1362	-0.945 to +1.217	0.068
price + capex	165,000	-0.1181	-1.064 to +0.828	0.047
ore, all four	2,478	-0.1023	-0.542 to +0.338	0.002
ore chemistry (AI pair)	338	+0.0356	-0.090 to +0.161	0.004
ore metallurgy (mass yields)	2,140	-0.3991	-0.807 to +0.008	0.000
dummy group (true EVPI = 0)	0	+0.0016	-0.051 to +0.054	0.000

Every interval straddles zero. The nested estimator's standard error on the all-nine group is 0.6142 MUSD, larger than any group value in the table, so this table also fails to support a ranking. Its value is the definitional control: the all-nine nested estimate (+0.1793) agrees with the closed-form full-information value (+0.4005) to within 0.2212 MUSD against a 3-standard-error tolerance of 1.8426, which confirms the estimator is unbiased and merely imprecise. An earlier version of that control used a flat 0.15 MUSD threshold and fired, reporting a correct estimator as broken; the tolerance is now the measured standard error.

One quantity escapes the noise entirely. Perfect information on all parameters jointly requires no nesting and can be formed per draw as $d_k = \max_a V(a, \theta_k) - V(a^*, \theta_k)$, which cancels the shared NPV variance:

EVPI(all) = 0.4030 +/- 0.0040 MUSD (COMPUTED, 200,000 draws x 4 seeds; seed spread 0.0042, ratio to within-run standard error 1.07, consistent). That is a 150-fold precision gain over the nested form. It equals 2.126 percent of the prior best value of 18.96 MUSD, and information binds on only 8.53 percent of draws.

This is a CEILING. No measurement of any subset can be worth more than perfect information on everything. It is the number the recommendation rests on, because it does not require resolving a ranking the estimators cannot deliver.

4.3 What that implies for the assay campaign

The ore assay campaign costs 2,478.00 USD per sample, built as follows.

Parameter	Cost, USD	Tag	Basis
AI mean	64.60	SOURCED	Actlabs 2026 schedule: RX1 prep 12.40 plus Code 4B2-Std fusion ICP-MS 52.20 at the 11-plus price
AI lot-to-lot sigma	273.40	DERIVED	Actlabs SOURCED unit prices, 12.40 plus 5 x 52.20; the replicate count of 5 lots is ASSUMED

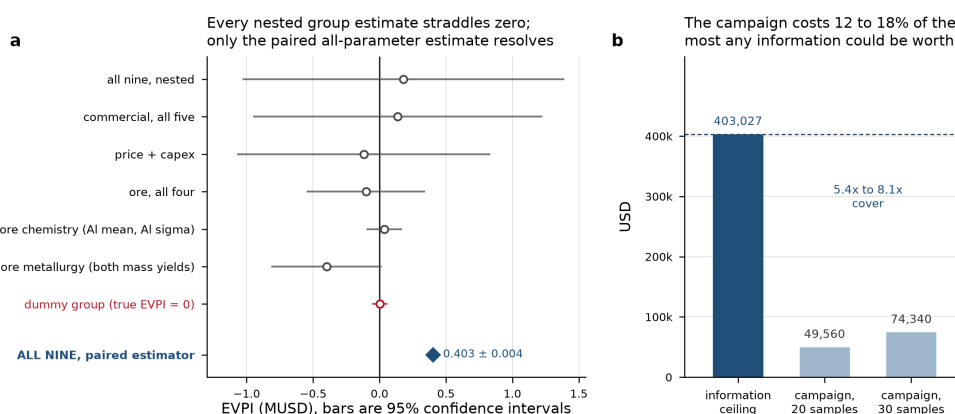
Parameter	Cost, USD	Tag	Basis
leach mass yield	1,140.00	DERIVED	Hazen July 2026 SOURCED prices: 6 x (leach 10 plus ICP-OES solids scan 155 plus 4-acid digestion 20) plus prep 30; the 6 leach conditions are ASSUMED
flotation mass yield	1,000.00	ASSUMED	proxy only: Hazen Bond ball mill grindability at 1,000 USD is SOURCED, but standing it in for a bench flotation test is judgement. Bench flotation is quoted By Quote at both laboratories

At 20 to 30 samples the campaign costs 49,560 to 74,340 USD, which is 12.3 to 18.4 percent of the 403,027 USD ceiling, a cover ratio of 5.4x to 8.1x. The sample count itself is ASSUMED: 20 to 30 samples labelled AE-Q-### comes from the project brief, not from any statistical power calculation performed here.

Recommendation: buy the ore assay campaign, and buy the AI chemistry first. The justification is deliberately not an EVPI ranking, since none is estimable. It is three separate arguments that agree:

- 1. Cost asymmetry.** The AI chemistry package costs 338 USD per sample against a 403,027 USD ceiling, a ratio of roughly 1,200 to 1. Even if its true EVPI is a small fraction of the ceiling, the expenditure is dominated.
- 2. Option value not in the EVPI number.** The AI result determines which product grades are reachable at all. That is a gate on the product map (HPQ crucible sand versus fused-grade filler versus silicon-metal feedstock), and the three-action model priced here does not represent product switching, so its EVPI understates the true value of the AI measurement.
- 3. Irreversibility.** AI chemistry is the one parameter no amount of process engineering can fix downstream, because lattice-bound Al, Ti, Li and B are not removable by acid leaching and set the ceiling grade of the deposit.

What NOT to buy first. The three parameters that dominate NPV variance are price, discount rate and capex, and none is an assay. Their group EVPI is +0.1362 MUSD against a 213,000 USD cost, a ratio below 1, so a market and EPC study is not justified at this stage on these numbers. This is the memo's sharpest practical result: **the parameters that dominate the variance are not the parameters worth measuring first**, because variance share says nothing about cost, and EVPI weighted by cost inverts the ranking.



Source: this study. (a) scripts/memo_evpi_joint.py, nested Monte Carlo at n_outer=512, n_inner=512, seed 11, no clamping at zero; the dummy group is a prior-drawn input the value function discards, so its true EVPI is exactly 0. Paired estimator: scripts/memo_evpi_ceiling.py, 200,000 draws x 4 seeds, forming max_a V - V(e*) per draw. (b) per-sample cost 2,478 USD is the sum of the four ore parameter costs (one SOURCED, two DERIVED from list prices, one ASSUMED proxy; see docs/memo_numbers.csv), an upper bound since shared sample preparation is not netted out.

Figure 4. (a) Group EVPI with 95 percent confidence intervals, all straddling zero, against the paired all-parameter estimate. (b) The information ceiling against campaign cost. Source: this study, scripts/memo_evpi_joint.py and scripts/memo_evpi_ceiling.py; Raiffa and Schlaifer 1961; Howard 1966 (doi:10.1109/TSSC.1966.300074).

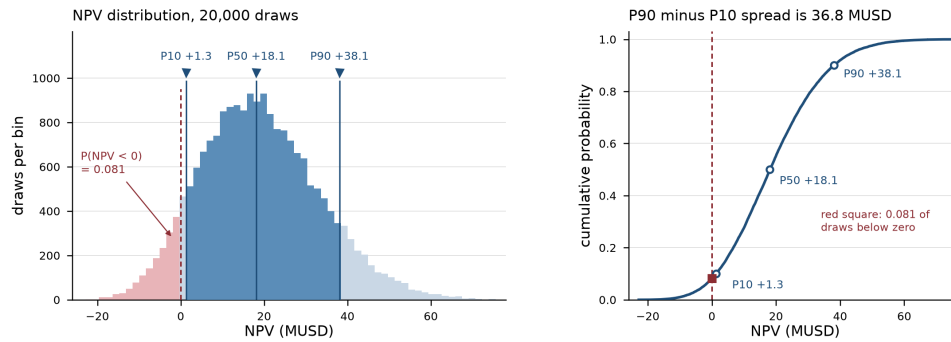
5. The NPV distribution, reported last and for a reason

20,000 draws, seed 42, zero failed evaluations (COMPUTED).

Quantity	P10	P50	P90	Spread
NPV, MUSD	+1.31	+18.11	+38.08	36.78
overall yield	0.6604	0.7346	0.8022	0.142
cash cost, USD/t	224.6	234.3	245.3	20.6
breakeven price, USD/t	2,279	2,807	3,511	1,232

Mean +19.05, standard deviation 14.31, range -22.82 to +90.05 MUSD. $P(\text{NPV} < 0) = 0.0810$. Bootstrap standard errors on the NPV percentiles (400 resamples) are P10 0.131, P50 0.117, P90 0.209 MUSD.

Read the P50 as arithmetic on ASSUMED bands, not as a forecast. The distribution appears here, after the sensitivity analysis, because the decision turns on which uncertainty dominates and what it costs to resolve, not on a point estimate. The breakeven price band is the more decision-relevant output: at a P90 breakeven of 3,511 USD/t the project needs a price near the upper half of its own ASSUMED 2,800 to 5,200 USD/t band, and that band is itself the least defensible input in the model.



Source: this study, `ae.econ.uncertainty.monte_carlo` over `plant_npv` in `tests/test_integration_uncertainty.py`, 20,000 draws, seed 42, zero failed evaluations. Bootstrap standard errors on the percentiles: P10 0.131, P50 0.117, P90 0.209 MUSD (400 resamples). This distribution is conditional on ASSUMED input bands. The Vikarabad deposit is uncharacterized in the citable record, so it is a scenario spread, not a forecast, and it is reported AFTER the sensitivity results deliberately: the decision turns on which uncertainty dominates, not on any point estimate here.

Figure 5. (a) NPV distribution with P10, P50 and P90 marked, draws below zero in red. (b) The same draws as a cumulative distribution. Source: this study, `scripts/memo_run.py`.

6. Assumption register

Parameter	Value or band	Tag	Basis
leach mass yield	0.80 to 0.97, mode 0.92	ASSUMED	scenario band, no Vikarabad assay exists
flotation mass yield	0.70 to 0.93, mode 0.85	ASSUMED	scenario band, no Vikarabad assay exists
AI mean	8 to 26 ppm, mode 18	ASSUMED	scenario band; 18 ppm is the mu that meets Cpk 1.33 at USL 30 ppm with sigma 3
AI lot-to-lot sigma	1.5 to 6.0 ppm, mode 3.0	ASSUMED	scenario band
HPQ price	2,800 to 5,200 USD/t, mode 3,500	ASSUMED	no transacted price sourced in this session
power price	0.045 to 0.095 USD/kWh, uniform	ASSUMED	no tariff sourced
reagent price	1.20 to 2.60 USD/kg, mode 1.80	ASSUMED	no quotation sourced
capex	25 to 60 MUSD, mode 38	ASSUMED	no engineering estimate sourced
discount rate	0.10 to 0.20, mode 0.14	ASSUMED	no financing terms established
AI upper spec limit	30 ppm	ASSUMED	scenario specification, set in the test module
Cpk target	1.33	SOURCED	Montgomery, Introduction to Statistical Quality Control, 7th ed., Wiley 2012, ch. 6; JEDEC JESD46
feed rate	7,000 t/period	ASSUMED	scenario scale
tax rate	0.21	ASSUMED	not tied to a sourced Indian rate
life	15 periods, 2 construction, ramp 0.35/0.75/1.00	ASSUMED	scenario schedule
polish capex, opex, AI factor	+9.0 MUSD, +35 USD/t, x0.60	ASSUMED	scenario stage, factor bounded in (0,1) on physical grounds
assay unit prices	see section 4.3	SOURCED / DERIVED / ASSUMED	Actlabs 2026-01-22 and Hazen July 2026 published fee schedules
sample count 20 to 30	20 to 30	ASSUMED	project brief, not a power calculation

Note on what is NOT sourced. Bench flotation testing and LA-ICP-MS are quoted "By Quote" or "by request" at both Actlabs and Hazen, so no list price exists to cite and the flotation cost is an ASSUMED proxy. The five commercial parameters (price, capex, power price, reagent price, discount rate) are not laboratory services at all; their "measurement costs" are ASSUMED study budgets with no citable list price. Capital cost scaling exponents: NOT SOURCED. A Crossref search for the six-tenths rule returned only off-topic results and then rate limited, so no scaling exponent is used anywhere in this analysis.

7. What would falsify the thesis

Each item is a measurement with a threshold that would change the recommendation.

- AI above the specification ceiling.** If the assay campaign returns a mean AI above roughly 26 ppm with lot-to-lot sigma above 6 ppm, the deposit cannot meet a 30 ppm USL at Cpk 1.33 by any process route, since lattice-bound AI is not leachable. HPQ crucible sand is off the table and only the lower-grade product map remains.
- Mass yield below the band.** With all other parameters at their modes, the combined flotation-times-leach mass yield that drives NPV to exactly zero is 0.5385 (COMPUTED by root-finding on `plant_npv`). Scaling both stages equally, that corresponds to leach 0.7634, which is BELOW its ASSUMED band low end of 0.80, and flotation 0.7053, which is INSIDE its ASSUMED band (0.70 to 0.93) by a margin of 0.005. So the honest statement is narrower than "yield must leave its band": the leach stage must fall below its band, while the flotation stage need only sit fractionally above its own worst case. Two comparisons matter for calibration. The Monte Carlo overall-yield P10 of 0.6604 maps to an NPV of +10.57 MUSD when everything else sits at its mode, not to the NPV P10. And the NPV P10 of +1.31 MUSD is a joint outcome: the 248 draws within 0.4 MUSD of it have a mean overall yield of 0.709, spanning 0.645 to 0.773. Yield and NPV percentiles do not correspond, because the two mass yields together carry only 0.1142 of NPV variance (correlation with NPV 0.359).
- Price below breakeven.** With all else at its mode, NPV crosses zero at a price of 2,466 USD/t (COMPUTED by root-finding), which is BELOW the low end of the ASSUMED 2,800 to 5,200 USD/t band. So price alone cannot sink the project inside its own band, and the relevant threshold is the Monte Carlo breakeven distribution instead: P50 2,807 and P90 3,511 USD/t. A transacted price below 2,807 USD/t makes the median scenario unbankable. This is the parameter with the largest variance share and the weakest sourcing, and no transacted HPQ price was sourced in this session.
- Capex above 62.4 MUSD.** NPV crosses zero at 62.44 MUSD of capex (COMPUTED), just above the 60 MUSD high end of the ASSUMED band. An engineering estimate above that inverts the decision on its own, and this is the one parameter whose adverse end nearly reaches zero without

help from any other.

5. **Discount rate above 0.2245.** NPV crosses zero at a discount rate of 0.2245 (COMPUTED), above the 0.20 high end of the ASSUMED band. Financing terms materially worse than 20 percent would be disqualifying on their own, which matters because the discount rate is the second largest variance contributor and is set by negotiation rather than measurement.
6. **Fluid inclusion or mineral inclusion density too high.** Not modelled here at all, and a gap in this analysis: inclusion density governs bubble formation in fused quartz and can disqualify ore that passes a trace-element assay. The characterization campaign should measure it even though the economic model has no term for it.
7. **The decision model itself.** If product switching (HPQ sand versus filler versus silicon-metal feedstock) were represented instead of the three-action set used here, the EVPI of the AI measurement would rise, possibly by a lot. The 0.4030 MUSD ceiling is a ceiling for THIS decision problem, not for the business.

8. Reproduction

```
cd ae-platform && export PYTHONPATH=src:tests:scripts
python scripts/memo_run.py           # MC, Sobol, tornado, headline EVPI
python scripts/memo_evpi_control.py  # three EVPI noise controls
python scripts/memo_evpi_joint.py    # group EVPI, definitional control
python scripts/memo_evpi_ceiling.py  # paired EVPI(all), the ceiling
python scripts/memo_append_csv.py    # control/group/ceiling rows into the CSV
python scripts/memo_falsification.py # section 7 thresholds, solved by root-finding
pytest tests/test_memo_numbers.py tests/test_memo_traceability.py
```

Outputs: docs/memo_numbers.csv (430 data rows, one per reported quantity, each with a tag and a basis note), docs/memo_numbers.json, docs/memo_evpi_control.json, docs/memo_evpi_joint.json, docs/memo_evpi_ceiling.json, docs/memo_falsification.json, and docs/figures/.

Measured test result, quoted from the run: 24 passed in 0.82s, being 9 in tests/test_memo_numbers.py and 15 in tests/test_memo_traceability.py. One of those 15 checks this very sentence: it counts the test functions in both files and fails if the number quoted here drifts from them, which it did once already.

The guards were verified by control in both directions, each time by reintroducing the defect, observing the named test fail, then restoring and observing it pass.

1. Restoring a blanket fee-schedule provenance note in the CSV writer made test_csv_written_by_the_script_carries_no_fabricated_provenance fail with "CSV row for price claims actlabs provenance" (1 failed, 8 passed).
2. Flipping the flotation cost tag back to DERIVED made test_flotation_cost_is_assumed_not_derived fail with assert 'DERIVED' == 'ASSUMED' (1 failed, 8 passed).
3. Stripping the appended sections from the CSV, which restores the file to exactly the 311-row state that made this memo's traceability claim false, made test_every_load_bearing_quantity_has_a_csv_row fail by naming all 21 missing (section, quantity) pairs, and test_group_evpi_rows_match_the_joint_json fail with "no CSV row for group ore_all_four".
4. Reintroducing the conflated yield threshold ("below about 0.66 puts the project at the modelled P10") made test_memo_does_not_claim_the_yield_p10_is_the_npv_p10 fail (1 failed, 8 passed).

After each restoration the suite returned to passing.

Errors made and corrected during this analysis, recorded deliberately

1. The first shipped docs/memo_numbers.csv attached a single Actlabs-and-Hazen provenance note to all nine measurement costs, including five that are not laboratory services. The background run had imported the pre-fix module. The CSV was regenerated from the saved JSON and now carries zero such rows.
2. This memo's own traceability claim was false when first written. The CSV came from scripts/memo_run.py alone, so it held the Monte Carlo, Sobol and tornado results but had no row for the paired ceiling (0.4030 MUSD), the group EVPI table, the null floor (0.1233 MUSD), the seed replication spread, or the campaign costs: precisely the numbers section 4 rests on. Fixed by scripts/memo_append_csv.py (111 rows) and scripts/memo_falsification.py (8 rows), and now guarded by tests/test_memo_traceability.py, which fails if any load-bearing quantity loses its row.
3. The group EVPI definitional control used a flat 0.15 MUSD tolerance and fired on a 0.2212 MUSD gap, reporting a correct estimator as broken. The tolerance is now three times the measured standard error of the estimate.
4. The sample count 20 to 30 was initially attributed to this memo's task brief. It comes from the project brief. The attribution is corrected in scripts/memo_evpi_ceiling.py and tagged ASSUMED here.
5. A draft caption asserted that the tornado and Sobol orderings "disagree on ranks 3 and 4". Checked against the data: they agree on ranks 1 through 7 and swap 8 and 9. Corrected before use.
6. Section 7 originally read a yield percentile as an NPV percentile, claiming a combined mass yield below 0.66 put the project at the NPV P10 of +1.31 MUSD. Those are unrelated quantities: 0.6604 is the overall-yield P10, and it maps to an NPV of +10.57 MUSD with everything else at its mode. The zero-NPV yield root is 0.5385, solved rather than asserted, and a test now blocks the old sentence from returning.
7. A draft response called figure 5 "clean" while the overlap checker was still printing two text-text pairs, and the checker that produced that output had been narrowed to exclude annotation objects, so it was not auditing all of the figure's text. Both problems are fixed. The figures were re-audited on the VECTOR PDFs, extracting every text run with its bounding box and testing all pairs at a 0.3 pt tolerance: figures 2, 3, 4 and 5 return no overlapping runs at all. Figure 1 returns one pair, "flo" against "flot. yield", which is a text-extraction artefact rather than a defect: the extractor splits a single rotated tick label into two runs whose rotated bounding boxes necessarily overlap each other. That panel's nine rotated labels were then inspected visually in two crops covering the full axis width (the first seven, then the last four, overlapping at AI mean and AI sigma) and all nine are separated. A first version of this paragraph claimed visual confirmation of all nine from a single crop that contained only seven, which is recorded here because the habit of overstating the scope of a check is the same error as overstating a result.